09/11/26

Signal Processing

- Recap: GLP systems, types I & IV ✓
- FIR filter design
  - Windowing ✓
  - Optimum Approximation: Type I filter
    - Problem formulation: A_e, H_d, W, E, ω_p, ω_s, M, δ_1, δ_2, min max, α_k
    - Alternation Theorem - F_p, P, D_p, W_p, E_p, $\|E\|$, uniqueness

- FIR filter design:

  - Windowing method

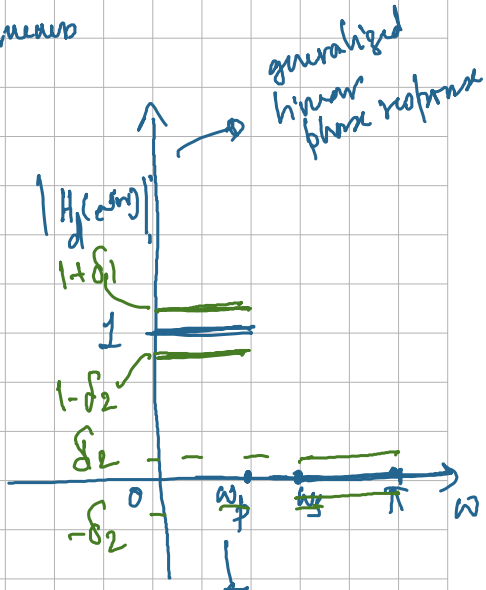
  - Let $h_d[n]$ be the desired impulse response.

  - Let $h[n]$ be the impulse response of the FIR filter of length $M+1$.

  - Pick a window $w[n]$ of length $(M+1)$ and multiply it with the desired impulse response $h_d[n]$, i.e.,

$$h[n] = w[n] \cdot h_d[n].$$

  - Example: $w[n] = \begin{cases} 1, & 0 \le n \le M \\ 0, & \text{elsewhere} \end{cases}$ → Rectangular window



  - The impact of windowing in the time domain is convolution in the frequency domain. i.e; $H(e^{j\omega}) = W(e^{j\omega}) * H_d(e^{j\omega})$

This leads to imperfections in the frequency domain which are directly related to $W(e^{j\omega})$

- Refer to Chap 7 in book for other window types. Bartlett, Hamming, Hanning etc.
- Note: Windows are symmetric to satisfy linear phase requirements

- **Optimal** FIR design: Type I FIR filter example. M is even
  - $h_e[n] = h_e[-n]$ even symmetry
  - $A_e(e^{j\omega}) = h_e[0] + \sum_{n=1}^{L} h_e[n] \cos(\omega n)$ $\quad L = M/2$
  - Let $\cos\omega n = T_n(\cos\omega)$ (Expressing $\cos\omega n$ as a polynomial of degree n).
    $T_n(\cdot)$ is a degree n polynomial. Chebyshev polynomial.
  - $A_e(e^{j\omega}) = \sum_{k=0}^{L} a_k (\cos\omega)^k$; Note that a_k's are related to h_e's.

    $= \sum_{k=0}^{L} a_k x^k$ where $x = \cos\omega$

    $= P(x)\Big|_{x=\cos\omega}$; $\left(P(x) = \sum_{k=0}^{L} a_k x^k\right)$
  - $E(\omega) = W(\omega)\left[H_d(e^{j\omega}) - A_e(e^{j\omega})\right]$;
    $W(\omega)$: weight function (non negative);
    $H_d(e^{j\omega})$: ideal or desired freq response: $H_d(e^{j\omega}) = \begin{cases} 1, & 0 \le \omega \le \omega_p \\ 0, & \omega_s \le \omega \le \pi \end{cases}$
    $A_e(e^{j\omega})$: Type I FIR freq response.

    $W(\omega) = \begin{cases} \frac{1}{K}, & 0 \le \omega \le \omega_p \\ 1, & \omega_s \le \omega \le \pi \end{cases}$ $\quad K = \frac{\delta_1}{\delta_2}$ $\quad K = \delta_1/\delta_2$

- Problem formulation: We want to find $h_e[n]$ (or alternatively a_k) such that we solve the following optimization

$$\min_{\{h_e[n]: 0 \le n \le L\}} \max_{\omega \in F} |E(\omega)| \qquad F = [0, \omega_p] \cup [\omega_s, \pi]$$

- Alternation theorem: Let F_p be a union of closed disjoint intervals of real numbers. Let $P(x) = \sum_{k=0}^{r} a_k x^k$, $(x \in F_p)$ be a polynomial of degree r defined on F_p. Define $D_p(x)$ to be the desired cont. function defined on F_p, and let $W_p(x)$ be a non-negative continuous function on F_p.

$$E_p(x) = W_p(x)\left[D_p(x) - P(x)\right], \text{ and let}$$

$$\|E\| = \max_{x \in F_p} |E_p(x)|$$

$P(x)$ is a unique degree r polynomial which minimize $\|E\|$ if there are $(r+2)$ alternations in F_p